

NOETICA MASTER BLUEPRINT

PART 06

Data Sources & Intelligence Collection Framework

Purpose

The purpose of the Data Sources & Intelligence Collection Framework is to provide Noetica with a continuously updating view of human knowledge.

Noetica cannot become a scientific intelligence system if it only observes papers.

Knowledge emerges through many channels:

- Research papers
- Patents
- Grants
- Conferences
- Datasets
- Open-source projects
- Startups
- Government programs
- Standards
- Clinical trials
- Scientific debates

Therefore Noetica must operate as a multi-source intelligence network.

Core Principle

No single source represents reality.

Every source is incomplete.

Noetica must synthesize evidence from multiple sources before evaluating significance.

Intelligence Collection Layers

Layer 1

Research Intelligence

Layer 2

Technology Intelligence

Layer 3

Funding Intelligence

Layer 4

Open Source Intelligence

Layer 5

Institutional Intelligence

Layer 6

Historical Intelligence

Layer 7

Weak Signal Intelligence

Layer 1

Research Intelligence

Purpose:

Track scientific discoveries.

Sources

arXiv

Purpose:

Early-stage research.

Fields:

- Mathematics
- Physics
- Computer Science
- Statistics
- Quantitative Biology
- Economics

Importance:

Detect breakthroughs before peer review.

Reliability:

Tier 3

PubMed

Purpose:

Biomedical research.

Fields:

- Medicine
- Biology
- Clinical Sciences
- Public Health

Importance:

Primary biomedical source.

Reliability:

Tier 1-2

OpenAlex

Purpose:

Global scholarly metadata.

Tracks:

- Papers
- Authors
- Institutions
- Citations
- Concepts

Importance:

Foundation discovery source.

Reliability:

Tier 1

CrossRef

Purpose:

DOI metadata and publication tracking.

Importance:

Discovery linkage.

Reliability:

Tier 1

Semantic Scholar

Purpose:

Citation networks.

Importance:

Research influence analysis.

Reliability:

Tier 1

bioRxiv

Purpose:

Biology preprints.

Importance:

Early biological discoveries.

Reliability:

Tier 3

medRxiv

Purpose:

Medical preprints.

Importance:

Early clinical signals.

Reliability:

Tier 3

Layer 2

Technology Intelligence

Purpose:

Track transformation of discoveries into technologies.

Patent Sources

USPTO

United States patents.

Tracks:

- New inventions
 - Technology evolution
 - Commercialization
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WIPO

International patent activity.

Tracks:

- Global innovation
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Google Patents

Patent discovery platform.

Tracks:

- Patent families
 - Inventor networks
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Patent Intelligence Signals

Patent Growth

Patent Citations

Patent Clusters

Technology Convergence

Commercialization Activity

Patent Families

Layer 3

Funding Intelligence

Purpose:

Track where future discoveries are likely to emerge.

Funding often predicts future scientific activity before publications appear.

Government Sources

NIH

Biomedical research funding.

NSF

Science and engineering funding.

ERC

European research funding.

Horizon Europe

Strategic scientific investment.

National Research Programs

Country-specific initiatives.

Examples:

AI Missions

Quantum Missions

Space Programs

Biotechnology Programs

Funding Signals

Grant Volume

Grant Growth

Funding Concentration

Emerging Research Programs

Strategic National Investments

Layer 4

Open Source Intelligence

Purpose:

Track technology adoption and implementation.

Many modern breakthroughs emerge through open-source ecosystems.

Sources

GitHub

Primary source.

Tracks:

Repositories

Contributors

Forks

Stars

Issues

Releases

Activity

GitLab

Additional source.

Hugging Face

Critical for AI.

Tracks:

Models

Datasets

Community Adoption

Open Source Signals

Repository Growth

Contributor Growth

Dependency Networks

Model Adoption

Community Development

Implementation Momentum

Layer 5

Institutional Intelligence

Purpose:

Understand who is producing knowledge.

Organizations

Universities

Research Institutes

Companies

Government Agencies

Think Tanks

Laboratories

Examples

MIT

Stanford

CERN

NIH

EMBL

DeepMind

OpenAI

Anthropic

NASA

ISRO

Institutional Signals

Discovery Output

Research Quality

Funding Activity

Collaboration Networks

Emerging Research Programs

Cross-Disciplinary Activity

Researcher Intelligence

Purpose:

Track individual contributors.

Metrics

Publication Activity

Discovery Contributions

Cross-Field Influence

Emerging Influence

Collaboration Networks

Innovation Velocity

Layer 6

Historical Intelligence

Purpose:

Provide context.

No discovery exists in isolation.

Historical Registry

Three Timelines

Foundational Discoveries

Question:

What changed civilization?

Examples:

- Zero
- Calculus
- Evolution
- Germ Theory

- Relativity
 - Quantum Mechanics
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Modern Breakthroughs

Question:

What changed science?

Examples:

- PCR
 - Internet
 - Human Genome
 - CRISPR
 - AlphaFold
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Emerging Frontiers

Question:

What may change the future?

Examples:

- AI Scientists
 - Quantum Networks
 - Synthetic Cells
 - Brain Computer Interfaces
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Historical Intelligence Functions

Lineage Mapping

Dependency Analysis

Knowledge Trees

Discovery Evolution

Civilizational Context

Layer 7

Weak Signal Intelligence

Purpose:

Detect important developments before mainstream recognition.

This is one of Noetica's most valuable capabilities.

Weak Signals

Characteristics:

Low attention

High novelty

Growing momentum

Increasing connectivity

Early-stage validation

Examples

Transformer Architectures (2017)

CRISPR (Early Years)

AlphaFold (Before Widespread Adoption)

Protein Language Models

Graph Neural Networks

Weak Signal Sources

Preprints

Workshops

Conference Posters

Research Blogs

Open Source Projects

Grant Announcements

Patent Filings

Small Research Communities

Conference Intelligence

Purpose:

Detect discoveries before publication.

Sources

NeurIPS

ICML

ICLR

AAAI

ACL

EMNLP

ISMB

RECOMB

APS Meetings

ACS Meetings

Nature Conferences

Science Conferences

Conference Signals

Accepted Papers

Workshops

Keynotes

Tutorials

Emerging Topics

Community Momentum

Dataset Intelligence

Purpose:

Track knowledge infrastructure.

Datasets often enable entire scientific revolutions.

Examples

Human Genome Project

ImageNet

AlphaFold Database

TCGA

Protein Data Bank

UK Biobank

Dataset Signals

Dataset Growth

Dataset Usage

Citation Activity

Research Dependency

Cross-Field Adoption

Startup Intelligence

Purpose:

Track commercialization signals.

Sources

Startup Databases

Funding Announcements

Accelerators

Incubators

VC Reports

Public Company Filings

Startup Signals

Funding Growth

Company Formation

Technology Adoption

Talent Migration

Market Creation

Government Intelligence

Purpose:

Track strategic priorities.

Governments often reveal future scientific directions before markets.

Signals

National Strategies

Research Missions

Defense Programs

Infrastructure Investments

Technology Roadmaps

Regulatory Changes

Source Reliability Framework

Every source receives a reliability tier.

Tier 1

Highest Confidence

Examples:

Peer-reviewed journals

Major institutions

Official databases

Government datasets

Tier 2

Strong Confidence

Conference proceedings

Established organizations

Tier 3

Moderate Confidence

Preprints

Unreviewed technical reports

Tier 4

Weak Confidence

Blogs

News articles

Opinion pieces

Tier 5

Minimal Confidence

Social media

Rumors

Unverified claims

Intelligence Fusion Layer

Noetica never relies on a single source.

Instead:

Paper

+ Patent

+ Grant

+ Dataset

+ Conference Activity

+ Open Source Adoption



Discovery Cluster



Scientific Significance Evaluation

Collection Frequency

Real-Time (Future)

Daily

Weekly

Monthly

Historical Backfills

Final Principle

The objective of the Data Sources & Intelligence Collection Framework is not to collect information.

The objective is to build a continuously updating evidence network that allows Noetica to identify meaningful discoveries, understand their context, measure their significance, and forecast their future influence across the entire human knowledge ecosystem.